

Team tryout problem set.

Eight olympiad problems across the main contest topics. Work through as many as you can. This set is how we get to know how you think.

HOW TO USE THIS SET

Read this first.

1. You are **not** expected to solve all eight. A few complete, well-written solutions tell us far more than eight rushed ones.
2. Write full solutions. Explain your reasoning, not just the answer. Partial progress and good ideas count.
3. Work on your own. Using the internet to look up a problem defeats the purpose; we want to see *your* thinking.
4. Write up your solutions (typed or neatly handwritten and scanned) and submit them with your application.

SUBMIT BY

Friday, August 28, 2026. Email your write-up as a single PDF to team@losgatosmathcircle.org, or attach it to the form on the team page.

THE PROBLEMS

Eight problems.

Topics are labeled. Two are marked *Challenge*, so save them for last.

1

ALGEBRA

Find all polynomials $P(x)$ with real coefficients such that

$$P(x)P(x+1) = P(x^2 + x + 1) \quad \text{for all real } x.$$

Hint: think about what the roots of P would have to satisfy.

2

NUMBER THEORY

Prove that for every integer n , the number $n^5 - n$ is divisible by 30.

3

COMBINATORICS

Each cell of a 3×7 grid is colored black or white. Prove that there exist two rows and two columns such that the four cells lying at their intersections all have the same color.

4

INEQUALITIES

Let a, b, c be positive real numbers. Prove that

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}.$$

5

GEOMETRY

CHALLENGE

Let ABC be a triangle with incenter I . Line AI meets the circumcircle of ABC again at M . Prove that

$$MB = MC = MI.$$

Hint: chase the inscribed angles at B and C .

6

FUNCTIONAL EQUATIONS

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that, for all real x and y ,

$$f(x + y) = f(x) + f(y) \quad \text{and} \quad f(xy) = f(x)f(y).$$

7

INVARIANTS

The numbers $1, 2, 3, \dots, 2026$ are written on a board. In one move you erase two numbers a and b and write their absolute difference $|a - b|$. After 2025 moves a single number remains. Prove that this number is odd.

8

NUMBER THEORY

CHALLENGE

Find all integer solutions (x, y, z) of the equation

$$x^2 + y^2 + z^2 = 2xyz.$$

Hint: consider the parity of x, y, z , and what it forces.

READY TO APPLY?

When your solutions are written up, head back to the [team page](#) to send in your application. Good luck, and enjoy the problems.
